

Total No. of Questions :6]

SEAT No. :

[Total No. of Pages :3

P119

APR. -16/BE/Insem. - 16

B.E. (Mechanical)

POWER PLANT ENGINEERING (402047)

(2012 Course) (Semester - II)

Time : 1 1/2 Hour]

[Max. Marks :30

Instructions to the candidates:

- 1) Answer Q 1 or Q2, Q3 or Q4, Q5 or Q6*
- 2) Neat diagrams must be drawn wherever necessary.*
- 3) Figures to the right side indicate full marks.*
- 4) Assume Suitable data if necessary*

Q1) a) Write a note on present status of power generation in India. **[5]**

b) Determine the annual cost of water softening plant from the following data:

Cost = Rs. 2.56×10^5

Salvage value = 6%

Life = 10 years

Annual cost of chemicals = Rs. 15000

Annual repair cost = Rs. 1000

Labour cost per month = Rs. 3000

Rate of interest by sinking fund method = 11% **[5]**

OR

Q2) a) Write short note on

i) Load Shedding

ii) Carbon Credits **[5]**

P.T.O.

b) The daily load for a power plant is given by the following equation:

$$L = 350 + 10 t - t^2$$

Where t is time in hours from 0 to 24 hrs and L in MW

i. Value of maximum load and when it occurs, and

ii. Plant load factor. [5]

Q3) a) What do you understand by coal beneficiation? [5]

b) The following observations were recorded during a test on surface condenser:

Condenser vacuum = 70 mm of Hg;

Barometer reading = 76.5 cm of Hg;

Mean condenser temperature = 35°C;

Hot well temperature = 28°C;

Condensate collected = 1800 kg/hr;

Air leakage = 1 kg/ ton of steam;

Cooling water circulated = 70 ton/ hr;

Cooling water inlet temperature 12°C;

Cooling water outlet temperature = 27°C.

Calculate

i. Vacuum efficiency

ii. Condenser efficiency [5]

OR

Q4) Short note on

- a) Down flow surface condenser and
- b) Central flow condenser [10]

Q5) a) Explain Mass Curve. [4]

- b) Explain with neat sketch The Boiling Water Reactor (BWR) Power Plant. [6]

OR

Q6) a) Short note on Hydrographs. [4]

- b) Short note on Nuclear Power Plants. State its advantages and disadvantages. [6]

